

8086 CALL and RET Instructions

1. CALL Instruction

Concept

The **CALL** instruction transfers program control to a **procedure (subroutine)**. Before transferring control, it automatically saves the **return address** on the **stack**, allowing execution to resume after the procedure finishes.

Rules / Notes

Key Concepts

- CALL changes program flow to a procedure
- Return address = instruction immediately after CALL
- Return address is stored on the stack
- Control returns using the **RET** instruction

2. Near CALL

Concept

A **Near CALL** transfers control to a procedure located **within the same code segment**.

Rules / Notes

Instruction Format

- Length: **3 bytes**
- Byte 1: Opcode
- Bytes 2–3: Signed displacement ($\pm 32K$)

Rules / Notes

Operation of Near CALL

1. Push offset address of next instruction onto stack
2. $IP \leftarrow IP + \text{displacement}$
3. Control transfers to procedure
4. RET pops saved offset into IP

Example

```
MAIN :  
    MOV AX , 1234H  
    CALL PROC1  
    MOV BX , AX  
  
PROC1 :  
    ADD AX , 0001H  
    RET
```

Exam Tip

Near CALL pushes **only IP** onto the stack. Code Segment (CS) remains unchanged.

3. Far CALL

Concept

A **Far CALL** transfers control to a procedure located **anywhere in memory**, possibly in a different code segment.

Rules / Notes

Instruction Format

- Length: **5 bytes**
- Bytes 2–3: New IP (offset)
- Bytes 4–5: New CS (segment)

Rules / Notes

Operation of Far CALL

1. Push current CS onto stack
2. Push current IP onto stack
3. Load new CS and IP
4. Transfer control to far procedure

Example

```
EXTRN FAR_PROC:FAR

MAIN:
    MOV AX, 1000H
    CALL FAR PTR FAR_PROC
    MOV BX, AX
```

Exam Tip

Far CALL pushes **CS and IP**. RET must restore both.

4. RET Instruction

Concept

The **RET** instruction returns control to the calling program by restoring the return address from the stack.

Rules / Notes

RET Behavior

- Near RET → pop IP
- Far RET → pop IP and CS

Comparison Summary

Feature	Near CALL	Far CALL
Procedure location	Same segment	Any segment
Instruction length	3 bytes	5 bytes
Stack push	IP	CS + IP
RET restores	IP	CS + IP

Exam Tip

Exam Shortcut:

- Near CALL ⇒ push IP
- Far CALL ⇒ push CS, then IP